

SATYAJIT ROY CHOUDHURY

Engineering Executive | 20+ years in US & India | Leader in Cloud, AI/ML Infra, and SRE Engineering
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EXECUTIVE SUMMARY

I build and scale engineering organizations that turn infrastructure investment into business outcomes. Over 20 years, I have run global platform and SRE teams through rapid growth, built reliability into products that couldn't afford to fail, and stayed close enough to the architecture to make the hard calls myself. I lead by developing the people around me, staying hands-on where it matters, and keeping complex systems honest with clear metrics and accountability.

KEY EXECUTIVE ACHIEVEMENTS

- **Hyperscale Platform Leadership:** Architected and operated search and AI platforms handling 30B+ daily requests at 99.98%+ availability across hybrid multi-cloud infrastructure, while increasing GPU cluster utilization by 38–40% through workload-aware scheduling.
- **Growth & Reliability Enablement:** Led platform and SRE strategy that absorbed 400% attack traffic growth and supported 200%+ SaaS customer expansion at F5 without proportional infrastructure cost increases while improving release velocity and incident response.
- **Financial & Cost Optimization:** Delivered 22–65% cloud cost reductions across three organizations through hybrid architectures, FinOps governance, and storage tiering including a 30% annual TCO reduction at F5 while accelerating feature delivery by 40%.
- **Organizational & Talent Leadership:** Built and led global engineering organizations up to 55+ engineers; maintained sub-10% attrition and advanced 30%+ of team members into senior or leadership roles through structured development and promotion planning.

AREAS OF EXPERTISE

Leadership & Strategy

- P&L & Budget Accountability
- Org Design & Headcount Planning
- Manager-of-Managers
- Build vs Buy Strategy
- Technical Roadmapping & OKRs
- FinOps / Cloud TCO Optimization
- Executive & Board Stakeholder Mgmt

Cloud & Platform Engineering

- Multi-Cloud (AWS, Azure, GCP)
- Kubernetes (EKS, AKS, GKE)
- GitOps (ArgoCD, Flux CD)
- IaC (Terraform, Crossplane)
- eBPF / Service Mesh (Cilium)
- Microservices & API Platforms
- CI/CD & Release Engineering

Reliability & Operations

- SLO/SLI Design & Incident Command
- MTTR Reduction (60–73%)
- Observability (OTel, Prometheus, ELK, Grafana, Jaeger)
- Chaos Engineering & Global On-call
- WAAP / WAF & Zero-Trust
- FedRAMP High · SOC 2 · PCI-DSS
- Mutual TLS & Policy-as-Code
- Runtime eBPF Monitoring

AI/ML & HPC Infrastructure

- NVIDIA GPU (V100, A100, T4)
- HPC Networking (RDMA, InfiniBand, RoCE)
- MLOps (Kubeflow, MLflow)
- Job Scheduling (Volcano)
- Inference Optimization MIG & GPU-aware Scheduling

Data & Storage Engineering

- Elasticsearch
- Object Storage Lifecycle
- High Ingestion Pipelines
- Multi-tenant Search Architecture
- Data Sovereignty & Compliance
- Sub-5ms P95 Latency Optimization

TECHNICAL EXPERIENCE

Engineering Leader at Mandolin – San Francisco, CA

March 2026 — Present

High-impact, hands-on leadership role (coach-leader type) driving the infrastructure and reliability function, focused on modernizing and scaling the core platform.

- Directed the SRE function, rapidly accelerating platform modernization. Integrated AI-driven tooling (Resolve.ai) to automate incident triage, achieving a **40% reduction in MTTR** and streamlining operations.
- Architected a zero-downtime migration to a multi-cluster GKE environment, significantly increasing **fault tolerance** and **system resilience**. Engineered scalable Kubernetes networking via Cilium/eBPF, deploying Robusta for rapid cluster troubleshooting. Led strategic database migration from MongoDB to Cloud SQL.
- Standardized end-to-end delivery using ArgoCD and Pulumi. Explored Pulumi Neo for AI-assisted IaC and enforced strict compliance guardrails with OPA/Kyverno to meet GDPR and HIPAA requirements.
- Established a FinOps framework via Kubecost and BigQuery, successfully driving a **53% reduction** in infrastructure costs while simultaneously supporting an expanding resource footprint.
- Hardened platform security with continuous vulnerability scanning (Wiz, Snyk) and robust secrets management. Designed scalable, secure access to internal cluster UIs via GCP IAP.
- Built the infrastructure team from inception. Accelerated developer velocity by launching internal portals (Cortex/Backstage) powered by Glean's AI enterprise search, reducing onboarding and knowledge discovery time by **over 50%**.

Director of Engineering and SRE at Arkose Labs – San Mateo, CA

Aug 2024 — June 2025

High-growth fraud detection SaaS. Owned platform engineering and SRE across reliability, cloud cost, and security posture while the platform absorbed sharp enterprise-driven traffic growth.

- Re-architected the platform to EKS-based microservices with an eBPF service mesh, supporting **7× transaction growth** over 10 months while holding **99.9% SLA** under sustained attack traffic.
- Cut P1 incident resolution time by **58%** by rebuilding observability and SOC signal quality, then leading high-severity incident command directly until the on-call team had the tools and process to own it.
- Drove migration from Cloudflare to CloudFront with Lambda@Edge, cutting edge latency by **35%** for enterprise customers.
- Built and executed FinOps governance program, reducing cloud spend by **22%** without limiting infrastructure capacity lowering effective cost-per-million transactions quarter over quarter.
- Established SLO-based release gates with Product, giving engineering and product a shared framework for trading feature velocity against reliability risk measurably reducing customer-impacting incidents.
- Ran build-vs-buy evaluations and vendor POCs across cloud, edge, and observability platforms selecting best-fit solutions that cut evaluation cycles and avoided long-term lock-in.
- Scaled the engineering organization from **4 to 16 engineers** across two teams; managed 2 engineering managers and owned the hiring plan and team structure.

Sr. Director of Product Engineering & Head of SRE at F5 Inc – San Jose, CA Oct 2022 — April 2024

Full P&L accountability for Platform Engineering and Global SRE for F5's **Distributed Cloud SaaS**, a globally distributed, security-critical platform spanning **25+ PoPs** across multiple cloud providers.

- Oversaw architecture strategy for a **25+ global PoP** platform delivering multi-cloud networking, WAAP/WAF, and edge services enabling the platform to absorb **400%** growth in attack traffic while supporting **200%+** customer expansion.
- Reduced annual **TCO by 30%** and accelerated feature delivery by **40%** through a disciplined build-vs-buy governance process that balanced internal IP development with the right vendor solutions.
- Partnered with the CISO to achieve FedRAMP High, PCI-DSS, and SOC 2 compliance — implementing policy-as-code, zero-trust architecture, mutual TLS, and runtime eBPF monitoring — unlocking regulated telecom and financial services markets.

- Cut MTTR by **73%** and sustained **99.92%+** platform availability by re-architecting observability with ELK, Jaeger, and ML-based anomaly detection.
- Rebuilt the global SRE model into a follow-the-sun structure, reducing on-call burnout, cutting attrition by 10 percentage points, and improving incident response consistency across time zones.
- Translated Board and C-suite priorities into multi-quarter engineering roadmaps and OKRs, aligning Cloud, SRE, and product teams across the full F5 Distributed Cloud portfolio.
- Led an organization of **55+** engineers as manager of managers: 3 Directors and 1 Senior Manager, owning hiring plans, headcount allocation, and operating budget.

Architect/Technical Leader at Adobe Inc – San Jose, CA

Aug 2018 — Oct 2022

Technical leader for Adobe's Core Search and Sensei Machine Learning platform hyperscale infrastructure serving billions of daily requests across the full Search and Sensei product ecosystem.

- Built and operated a hybrid **GPU/CPU infrastructure** across AWS, Azure, and on-prem that handled **~30B** daily API requests at **99.98%+** availability.
- Designed and optimized NVIDIA V100/A100 HPC clusters with RDMA InfiniBand, MIG, and GPU-aware scheduling (Kubeflow, Volcano), increasing cluster utilization by **38%** measurably reducing compute cost for equivalent workload volume.
- Migrated from managed AWS Elasticsearch to a self-managed hybrid architecture across 18 clusters handling **10B+** documents at **6,000 writes/sec**, delivering **30%** licensing savings while improving operational control.
- Consolidated fragmented Kubernetes environments into 15 multi-tenant clusters using Cilium (eBPF), eliminating **90%** of cluster sprawl; the reference architecture was later adopted across Adobe's broader engineering organization.
- Cut cloud storage costs by **65%** through data lifecycle and tiering policies across search and ML pipelines while maintaining **sub-5ms P95** latency.
- Led 12 senior platform and infrastructure engineers, setting the architectural direction for search and ML infrastructure used across Adobe's product portfolio.

Tech Leader at Macys.com – San Francisco, CA

Nov 2016 — Aug 2018

Technical leader for enterprise-wide CI/CD and platform modernization supporting Macy's and Bloomingdale's e-commerce engineering organizations.

- Designed and delivered a company-wide CI/CD platform using Jenkins pipelines, Spinnaker, and Kubernetes reducing deployment time from 2–3 days to under 1 hour and enabling 100+ tested deployments per week.
- Implemented blue-green and canary deployment strategies, enabling near-zero-downtime releases for revenue-critical e-commerce workloads during peak retail periods.
- Designed a hybrid cloud architecture across GCP (GKE), AWS (ECS/EKS), and on-prem VMware Tanzu, enabling consistent workload execution across environments.
- Introduced early GitOps workflows using Flux CD, improving deployment consistency and reducing infrastructure provisioning errors during peak retail traffic.
- Led a team of 15 engineers across platform modernization and delivery tooling for both the Macy's and Bloomingdale's engineering organizations.

EARLIER EXPERIENCE

- **Prior Engineering Roles (2009 – 2016):** Workday, Chegg, RocketFuel, Adobe, Saba Software, CMC
- **International Engineering Roles (2004 – 2009):** Autonomy, SDG Group, HCL Tech, vCustomer (India)

EDUCATION

- **Bachelor of Science** | Mahatma Gandhi Kashi Vidyapith | Varanasi, UP, India | June 2002
- **Leading Effective Decision-Making (certification)** | Yale School of Management | Online | October 2021